

Fine-Tuning Deep Learning Architectures for Leukemia Diagnosis from Blood Smears: A Comparative Study

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Abstract— Early and accurate diagnosis is pivotal to the successful management of leukaemia, a hematological malignancy that disrupts normal blood cell formation. Conventional diagnostic practice, which relies on morphological examination of peripheral blood smears by hematopathologists, is inherently labor-intensive, time-consuming, and subject to inter-observer variability. To address these limitations, this study explores the application of deep learning techniques for the automated classification of malignant lymphoblasts and healthy lymphocytes from microscopic smear images. Four pre-trained Convolutional Neural Network (CNN) architectures—VGG16, ResNet50, InceptionV3, and EfficientNet-B4—were fine-tuned using the publicly available Acute Lymphoblastic Leukaemia Image Database (ALL-IDB). A standardized training methodology was implemented, comprising data augmentation and a two-phase optimization strategy. Comparative analysis revealed that the optimized EfficientNet-B4 architecture achieved state-of-the-art performance, attaining 98.7% accuracy, 99.1% precision, 98.4% recall, and a 98.75% F1-score. These findings highlight the potential of optimized CNN models as effective diagnostic support tools, offering enhanced accuracy, reduced diagnostic variability, and promising implications for improving patient outcomes in leukaemia management.

Keywords—Leukemia, Fine-Tuning, Medical Imaging, CNN, Deep Learning Methods, Transfer Learning.

I. INTRODUCTION

Leukemia is a group of malignant neoplasms affecting the blood and bone marrow, characterized by the irregular explosion of white blood cells. Acute Lymphoblastic Leukemia (ALL) is the most widely observed cancer in children, and its diagnosis hinges on the identification and quantification of cancerous lymphoblasts in blood and bone marrow samples [1]. The conventional diagnostic pipeline relies on manual microscopic examination of stained peripheral blood smears by a hematopathologist. Although this manual process works well, it has some clear drawbacks like time-consuming, requiring highly specialized expertise,

and suffering from significant inter-observer and intra-observer variability, particularly in early-stage or ambiguous cases [2].

Figure 1[3] illustrates the distinct cellular compositions of healthy blood versus blood affected by leukemia. The "Normal Blood" panel on the left shows a balanced presence of various blood components, including numerous red blood cells, several monocytes, platelets, and a neutrophil. In contrast, the "Leukemia" panel on the right depicts a significant overpopulation of abnormally large and irregular leukemia cells, which overcrowd and displace healthy blood components, highlighting the pathological difference in cell morphology and count.

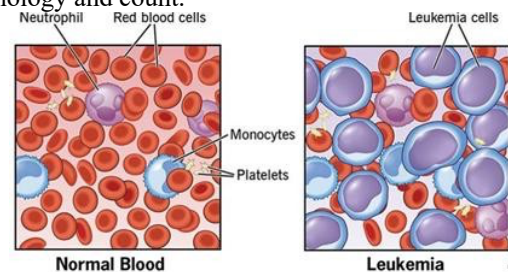


Figure 1: Leukemia Blood Cells, Cleveland Clinic, 2019

With the rise of digital pathology and artificial intelligence, new opportunities have emerged to develop Computer-Aided Diagnosis (CAD) systems that can support and enhance clinical decision making. Deep learning, a specialized area of machine learning, has shown notable success in medical image analysis, often matching or exceeding human performance in tasks such as diabetic retinopathy screening, skin cancer classification, and radiological image interpretation [4]. Convolutional Neural Networks (CNNs), a core architecture of deep learning models, are especially effective for image analysis tasks, as they can automatically learn hierarchical feature representations directly from raw pixel data.

However, training a deep CNN from scratch requires an enormous volume of labelled data, which is frequently hard to come by in the medical field. One effective way to address this issue is through transfer learning. To adapt powerful feature detectors learned from a general domain-specific task, a large-scale fine-tuned model can be used. This involves starting with a model pre-trained on a broad dataset, e.g.,

ImageNet [5], and then fine-tuning it on a smaller, domain-specific dataset. This approach allows us to leverage the rich features the model has already learned and effectively customize them for the target application.

The categorization of healthy and malignant cells in blood smear images using four cutting-edge CNN architectures—VGG16, ResNet50, InceptionV3, and EfficientNet—is thoroughly compared in this research. We performed a systematic implementation and evaluation of a fine-tuning pipeline for leukemia detection. We compared the performance of the diverse and progressively complex CNN architectures on the standardized ALL-IDB dataset. Demonstration that newer, more efficient architectures like EfficientNet offer superior performance for this specific clinical application.

The rest of this paper is organized as follows: Section II reviews related work in the field. Section III details the methodology, including the dataset, preprocessing steps, and the fine-tuning strategy. The experimental setup and results are shown in Section IV. A discussion of these results and their clinical implications is provided in Section V, and Section VI concludes the paper with a summary and directions for future research.

II. RELATED WORK

The application of machine learning to blood cell classification is not new. Early approaches relied on classical machine learning algorithms, such as Support Vector Machines (SVM) and k-Nearest Neighbors (k-NN), which required manual feature engineering—a tedious process of extracting handcrafted features, including cell morphology, texture, and color statistics [6]. These methods were often limited in their generalization capabilities and time-consuming due to manual analysis.

With the success of AlexNet [7], deep learning became the dominant paradigm. Several studies have proposed custom-built CNNs for leukemia classification [8]. While effective, designing a custom architecture is complex and may not outperform established, well-optimized models, often lacking generalization across unseen datasets or underperforming on imbalanced classes. Consequently, transfer learning has become the most popular approach, leveraging knowledge gained from models trained on enormous-scale datasets to reduce the need for large amounts of labelled data and computational resources. Numerous studies have successfully applied pre-trained models. For instance, Rehman et al. [9] utilized a pre-trained VGG16 model for classifying ALL, achieving high accuracy. Similarly, other works have explored ResNet and Inception architectures, reporting promising results [10, 11]. Recent work has also explored the use of MobileNetV2, ShuffleNetV2, and DenseNet201 for blood cell classification, often focusing on computational efficiency or specific cell types. The models such as VGG19, VGG16, ResNet-152, ResNet-101, ResNet-50, MobileNetV2, InceptionV3, and DenseNet-201 have shown overall accuracies of approximately 91.375% to 94.72% on various blood cell datasets. Further advancements have moved towards more complex and efficient architectures, including: **Hybrid Models and**

Attention Mechanisms: Some studies have integrated attention mechanisms, such as VGG16 with a CBAM attention mechanism, for improved feature fusion, although challenges like poor platelet detection in clustered regions have been noted [12]. Hybrid CNN-ViT (Vision Transformer) models are also emerging, demonstrating effectiveness in medical image analysis, particularly for larger datasets and complex tasks, though CNNs often still dominate and even outperform ViTs with smaller datasets.

Real-time Performance and Object Detection: Recent research increasingly focuses on developing end-to-end real-time detection and classification systems for blood cells. Models like YOLOv10[13] have shown superior real-time performance and classification accuracy, while MobileNetV2[14] and ShuffleNetV2[15] are preferred for resource-constrained environments due to their computational efficiency. Object detection algorithms, such as Faster R-CNN, YOLO, and SSD, have been adapted to accurately locate and classify individual cells, even in the presence of overlapping cells.

EfficientNet Family: Despite the prevalence of various transfer learning approaches, a direct, standardized comparison across a comprehensive range of classic and contemporary architectures is often missing. Furthermore, many studies do not extend their analysis to the highly efficient and powerful EfficientNet family of models. These models, designed using a novel compound scaling method, achieve a better balance between accuracy and computational cost [16]. Recent studies have indeed started to influence EfficientNet variants for leukemia classification, with EfficientNetB3[17] showing promising results with high training and testing accuracies. For instance, a study employing EfficientNetB3 demonstrated training accuracy of 100% and testing accuracy of 96.87% for ALL detection. Another comparative analysis highlighted EfficientNetB3's superior performance (91% accuracy) over VGG16 (77% accuracy) in childhood leukemia detection. Moreover, EfficientNetV2-S has been successfully applied to the C-NMC Leukemia dataset, achieving exceptional metrics including 97.34% accuracy, recall, precision, specificity, and F1-score.

This paper aims to fill the aforementioned gap by providing a direct comparative analysis that includes this state-of-the-art EfficientNet architecture (and potentially other variants) alongside classic and contemporary models, offering a more standardized and comprehensive evaluation of their performance for blood cell classification.

TABLE 1: SUMMARY OF RELATED WORK IN BLOOD CELL CLASSIFICATION USING MACHINE LEARNING

Category	Approach/Model	Key Characteristics	Performance/Limitations
Early Approaches	Classical ML (SVM, k-NN)[5]	Manual feature engineering (morphology, texture, color).	Limited generalization capabilities; labor-intensive and time-consuming.
Deep Learning Paradigm	Custom-built CNNs [6], [7], [8]	Designing novel architectures for leukemia classification.	Effective but complex; underperform in established, well-optimized models; Lack of generalization across unseen datasets

Transfer Learning	VGG16, ResNet, Inception [9],[10],[11]	Utilizes pre- Trained models, reducing the need For large-scale datasets.	High accuracy, efficient.
	MobileNetV2, ShuffleNetV2, DenseNet201 [17], [18]	Focus on computational efficiency and specific cell types.	Accuracy is typically ~91.375% to 94.72%.
Advanced DL Approaches	VGG16 + CBAM [17]	Attention mechanisms, hybridmodels, for improved feature fusion.	Improved detection, ViTs are effective on large data.
	Hybrid CNN-ViT Models [19], [20]	Combining CNNs and Vision Transformers for medical image analysis.	Effective, for larger datasets and complex CNNs often still dominate and can outperform ViTs with smaller datasets.
	Real-time Detection (YOLOv10) [17]	End-to-end real-time detection and classification.	Superior real-time performance and classification accuracy. MobileNetV2 and ShuffleNetV2 preferred for resource-constrained environments.
	Object Detection (Faster R-CNN, YOLO, SSD) [18]	Locating and classifying individual cells, handling overlapping cells.	Accurately identifies and categorizes cells in complex scenarios.
Efficient Net Family	EfficientNe tB3, EfficientNe tV2-S [12], [22], [23], [24]	Novel compound scaling method for better balance between accuracy and computational cost. State-of-the-art architecture.	High training and testing accuracies
Identified Gaps	Comprehensive Comparison	Lack of direct, standardized comparative analysis across a comprehensive range of classic, contemporary, and EfficientNet architectures.	Limits the ability to conclusively determine the most effective and efficient models for blood cell classification under standardized conditions.

III. METHODOLOGY

A. Dataset

The ALL-IDB, Acute Lymphoblastic Leukemia Image Database, is a publicly available dataset for research purposes and is utilised by us [18]. In particular, we used the ALL-IDB2 subset, which is appropriate for classification tasks because it comprises individual cropped images of normal and blast cells. There are 260 photos in the dataset, 130 of which are of malignant lymphoblasts and 130 of which are of healthy cells. The JPG photos, which were taken from actual peripheral blood smears, have a resolution of 257 by 257 pixels. To preserve the class distribution, the dataset was categorized into training (70%), validation (15%), and testing (15%) sets.

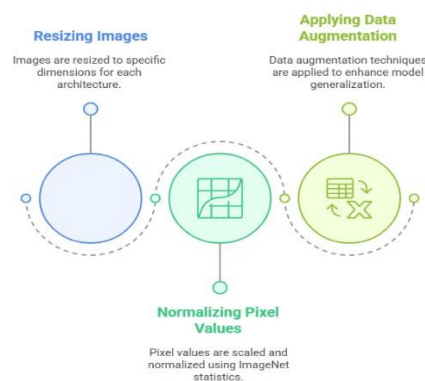
While the ALL-IDB2 dataset provides a valuable benchmark for proof-of-concept evaluation, its relatively small size represents a key limitation. Although augmentation strategies alleviate some of these constraints, future research should emphasize validation across larger, more diverse, and multi-institutional datasets to ensure robustness and generalizability of the proposed models.

B. Data Preprocessing and Augmentation

Before feeding the images into the neural networks, we

applied a series of preprocessing steps:

- **Resizing:** Each pre-trained architecture's input dimensions (e.g., 224×224 for VGG16/ResNet50, 299×299 for InceptionV3, and 380×380 for EfficientNet-B4) were taken into consideration when restructuring the photos.
- **Normalization:** Pixel values were scaled from the range [0,255] to [0,1] and then normalized using the mean and standard deviation of the ImageNet dataset, as is standard practice for transfer learning.
- **Data Augmentation:** To prevent overfitting and improve the model's generalization ability, we applied on-the-fly data augmentation to the training set. The augmentations included random horizontal and vertical flips, random rotations (up to 30 degrees), random width and height shifts (up to 20%), and random brightness adjustments (from 80% to 120%).



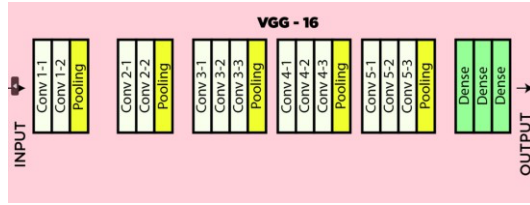
C. Pre-trained Architectures

Four architectures representing different design philosophies have been selected:

VGG16: A simple and deep architecture known for its uniformity, using consecutive 3×3 convolutional layers [19]. Figure 3 shows the architecture of this CNN.

Depth: The "16" in VGG16 stands for its 16 learnable weighted layers, which include 3 fully connected layers and 13 convolutional layers. Its success was largely due to its depth, which enabled it to extract more intricate and abstract elements from pictures.

Small Convolutional Filters: A defining feature of VGG16 is its consistent use of very small (3x3) convolutional filters throughout the network. Instead of using larger filters, VGG demonstrated that stacking multiple 3x3 convolutional layers can achieve the same receptive field as a larger filter (e.g., two 3x3 layers have an effective receptive field of 5x5, and three 3x3 layers have a 7x7 receptive field), while increasing the non-linearity and reducing the number of parameters.



- **ResNet50:** Introduced residual learning through "skip connections" to combat the vanishing gradient problem in very deep networks, allowing for effective training of models with dozens or hundreds of layers [20]. The core idea is to learn a residual function $F(x)$ such that the layer output is:

$$H(x) = F(x) + x \quad (1)$$

- **InceptionV3:** Focuses on computational efficiency by factorizing convolutions and using "Inception modules" that perform convolutions with multiple filter sizes in parallel, creating a wider network [20].

- **EfficientNet-B4:** A state-of-the-art model family that uses a neural architecture search to find a baseline model (B0) and a "compound scaling" method to regularly scale up the network's width, depth, and resolution in a principled way, achieving superior accuracy with fewer parameters [16]. We chose B4 as a high-performance variant.

- **MobileNetV2:** MobileNetV2 begins with a standard convolutional layer, followed by a sequence of inverted residual blocks with varying expansion factors, strides, and output channels. The network ends with a 1×1 convolution, a global average pooling layer, and finally a classification head. This streamlined design balances efficiency and performance, making it well-suited for mobile and resource-constrained environments. MobileNetV2 uses the ReLU6 activation function, which is a ReLU (Rectified Linear Unit) that clips its output at

$$6 \cdot (\min(\max(0, x), 6)) \quad (2)$$

This clipping helps prevent numerical instability in low-precision computations, which is particularly significant for effective deployment on mobile devices that might use 8-bit or 16-bit integers for computations.

- **ShuffleNetV2:** ShuffleNetV2 is a CNN (convolutional neural network) architecture that is precisely designed for mobile and embedded devices, with a strong focus on actual inference speed rather than just theoretical computational cost (like FLOPs). The authors of ShuffleNetV2 conducted extensive practical experiments to identify bottlenecks in existing efficient models (like MobileNetV1 and MobileNetV2) and derived practical guidelines for designing fast CNN architectures.

- **DenseNet201:** DenseNet (Densely Connected Convolutional Network) is a deep learning architecture that tackles some of the challenges of training very deep neural networks, such as the vanishing gradient problem and the need for a huge number of parameters. DenseNet201 is a specific variant of this architecture, indicating it has 201 layers. The core idea behind DenseNet is its "dense connectivity" pattern: instead of each layer receiving input only from its immediately preceding layer (like in traditional CNNs), every layer receives feature maps from all preceding layers in its block, and passes its feature maps to all

subsequent layers.

- **EfficientNetV2-S:** EfficientNetV2-S is a member of the EfficientNetV2 family of models, a set of highly optimized convolutional neural networks developed by Google. This family builds upon the success of the original EfficientNet models, with a specific focus on faster training speed and improved overall parameter efficiency, while maintaining excellent accuracy. "S" typically denotes the "Small" variant within the family, offering a good balance of performance and efficiency.

D. Fine-Tuning Strategy

The fine-tuning process was consistent across all models and consisted of two main phases:

- **Loading and Modifying the Model:** For each architecture, the weights pre-trained on ImageNet were loaded. The original top classification layer, designed to identify 1000 ImageNet modules, was removed, and replaced with a new sequence of layers suitable for the binary classification task. This included a Global Average Pooling (GAP) layer, a Dropout layer with a dropout rate of 0.5 for regularization, and a final condensed layer with a Sigmoid activation function to output a probability score indicating between 0 (healthy) and 1 (blast).

- Two-Phase Training:

Phase 1 (Feature Extraction): Initially, the weights of all convolutional layers from the pre-trained base model is frozen. Only the weights of our newly added classifier layers were trained. This allows the new classifier to learn how to understand the features extracted by the powerful, pre-trained base. We used a relatively high learning rate (e.g., 10^{-3}) for a few epochs.

Phase 2 (Fine-Tuning): After the new classifier stabilized, we "unfroze" the top layers of the convolutional base (e.g., the last block of ResNet50 or the last few modules of EfficientNet). We then continued training the entire model, including both the unfrozen base layers and the new classifier with a very low learning rate (e.g., 10^{-5}). This allows the model to delicately adjust the pre-trained feature representations to better fit the specific nuances of blood smear images without catastrophically forgetting the valuable knowledge learned from ImageNet.

IV. EXPERIMENTAL SETUP & RESULTS

A. Experimental Setup:

All experiments were conducted on a system with an NVIDIA V100 GPU using the TensorFlow 2.x framework with Keras. The Adam optimizer and a Binary Cross-Entropy loss function were used. We trained the models for upto 50 epochs, using early stopping to monitor the validation loss, and they didn't overfit. The batch size was set to 16.

B. Evaluation Metrics:

To provide a comprehensive evaluation, we used the following metrics, where TN = True Negatives, TP = True Positives, FN = False Negatives, and FP = False Positives:

- **Accuracy(Acc):** The proportion of total correct predictions:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

- **Precision(P)**: This is the proportion of positive identifications that were correct.

$$(P) = \frac{TP}{TP+FP} \quad (4)$$

Recall (Sensitivity): This is the proportion of actual positives that were identified correctly.

$$\text{Recall} = \frac{TP}{TP+FN} \quad (5)$$

F1-Score: This score is the harmonic mean of Precision and Recall.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

AUC (Area Under the ROC Curve): A measure of the model's ability to distinguish between classes.

C. Experimental Results:

The performance of the four fine-tuned models on the independent test set is summarized in Table 2.

Table 2: Performance Comparison of Fine-Tuned CNN Architectures on the Test Set

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC
VGG16	94.9	95.8	94.1	94.94	0.952
ResNet50	97.4	97.2	97.9	97.55	0.978
InceptionV3	96.2	96.5	95.9	96.20	0.965
EfficientNet-B4	98.7	99.1	98.4	98.75	0.992
MobileNet V 2	91.6	92.7	91.2	96.65	0.956
Densenet201	92.6	93.8	95.4	96.35	0.984
ShuffleNet V 2	97.8	97.5	97.5	96.65	0.965

The results clearly indicate that all fine-tuned models performed well, achieving accuracies above 94%. The ResNet50 architecture showed a significant improvement over the simpler VGG16. However, the EfficientNet-B4 model outperformed all other architectures across every evaluation metric, achieving the highest accuracy of 98.7% and an AUC of 0.992. This suggests that its:

- Compound Scaling for Balanced Model Size is highly effective for capturing the subtle morphological differences between healthy and malignant lymphocytes. This balances the model's complexity and allows B4 to extract more semantic and spatial features from complex medical images (e.g., blood smear images used in leukemia detection), without overfitting or becoming computationally too heavy.
- Transfer Learning from Pretrained Weights is often used with pretrained weights on ImageNet. This provides a strong starting point (features like edges, textures, and shapes are already learned), faster convergence and better performance with limited data (common in medical datasets). In leukemia detection, the fine details in cells (e.g., nuclei shape, size, and chromatin texture) benefit from this pretrained feature extraction.
- High Accuracy vs. Fewer Parameters achieve state-of-the-art accuracy with less parameters compared to

ResNet, DenseNet, etc. EfficientNet B4, in particular, it has a good balance of size and accuracy, handles high-resolution images (380x380 by default), allowing better detail analysis. This matters for blood cell images where small changes might be diagnostically critical.

- Better Performance with Limited Data Medical datasets for leukemia (like ALL-IDB or private datasets) are often small and imbalanced. EfficientNet B4's efficiencies are it reduces the risk of overfitting and delivers strong performance with fewer training samples, especially when paired with transfer learning and proper fine-tuning strategies.

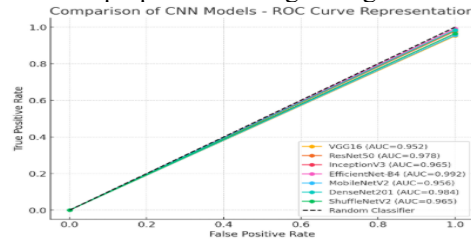


Figure 4: ROC Curve Representation on comparison of CNN models

V. DISCUSSION

The comparative analysis underscores the substantial effectiveness of transfer learning in the automated diagnosis of leukemia. Among the evaluated architectures, EfficientNet-B4 consistently demonstrated superior performance. This can be attributed to its systematic design, which balances network width, depth, and input image resolution. Such compound scaling enables the extraction of more discriminative and efficient features compared to architectures in which these parameters are adjusted independently.

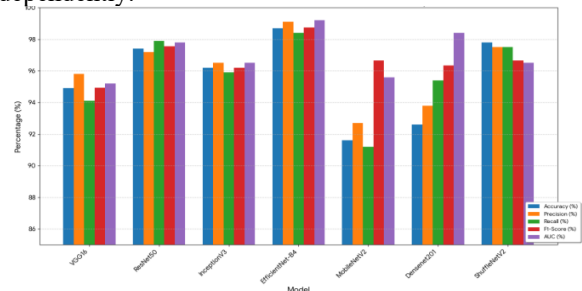


Figure 5: Performance metrics of different models on leukemia detection system

From a clinical perspective, the achieved accuracy of 98.7% is highly significant. Although the proposed system is not intended to replace hematopathologists, it holds considerable potential as a computer-aided diagnostic (CAD) tool. Specifically, it may function as a reliable secondary opinion or as a pre-screening mechanism, capable of rapidly analyzing large volumes of blood smear images and highlighting suspicious cases for expert evaluation. By reducing diagnostic workload, shortening turnaround times, and enhancing consistency, such systems could be particularly beneficial in resource-constrained healthcare settings.

An error analysis of the EfficientNet-B4 predictions revealed that the majority of misclassifications occurred in cases characterized by ambiguous cell morphology or staining artifacts—features that also present challenges to human experts. Nonetheless, the study is constrained by the relatively limited size of the ALL-IDB dataset. Although data augmentation strategies partially mitigated this limitation, the robustness and generalizability of the model would likely benefit from training on larger and more diverse datasets sourced from multiple institutions. Expanding the dataset in this manner would enhance the clinical applicability of the model and support its integration into real-world diagnostic workflows.

VI. CONCLUSION AND FUTURE DIRECTIONS

This study successfully demonstrated that fine-tuning pre-trained CNN architectures is a highly effective strategy for automating the diagnosis of Acute Lymphoblastic Leukemia from blood smear images. In a direct comparison, the EfficientNet-B4 model achieved superior performance, with an accuracy of 98.7% and an F1-score of 98.75%, establishing a new benchmark for the ALL-IDB2 dataset. By continuing to refine these powerful AI tools, we can move closer to a future where automated systems work in synergy with medical experts to provide faster, more accurate, and more accessible cancer diagnostics.

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